


$$\int_0^1 (1-x^2) dx < \frac{4n^2+3n-1}{6n^2}. \quad [4]$$
[illegible]

[illegible]

where a and b are constants.

- Show that $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = \frac{25}{4}(t^2 + t^{-2})^2$ and find the exact length of C . [6]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

- (b)** It is given instead that $a = b = 0$.

Find the value of $\frac{d^2y}{dx^2}$ when $t = 1$.

[4]

This image shows a full page of a document template designed for handwriting practice or general writing. It consists of approximately 28 evenly spaced horizontal dotted lines running across the width of the page. The background is plain white, and there are no margins, headers, footers, or other markings present.

- 8 (a) Starting from the definitions of $\tanh$ and sech in terms of exponentials, prove that

$$1 - \tanh^2 x = \operatorname{sech}^2 x. \quad [3]$$

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- (b) Using the substitution $u = \tanh x$, or otherwise, find $\int \operatorname{sech}^2 x \tanh^2 x \, dx$. [2]

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It is given that, for $n \geq 0$, $I_n = \int_0^{\ln 3} \operatorname{sech}^n x \tanh^2 x \, dx$.

- (c) Show that, for $n \geq 2$,

$$(n+1)I_n = \left(\frac{4}{5}\right)^3 \left(\frac{3}{5}\right)^{n-2} + (n-2)I_{n-2}. \quad [5]$$

[You may use the result that $\frac{d}{dx}(\operatorname{sech} x) = -\tanh x \operatorname{sech} x$.]

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(d) Find the value of I_4 . [3]

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